

CHAPTER 1

INTRODUCTION

The Introduction provides the reader with an overview of the proposed research and establishes the foundation for the discussions presented in Chapter 1. It briefly introduces the research domain, explains the significance of the study, and presents the overall organization of the chapter. Rather than discussing the research problem in detail, the Introduction prepares the reader for the succeeding sections by providing sufficient context to understand the direction and purpose of the study.

For Computer Science and Information Technology research, the Introduction should briefly mention the emerging technologies or application domain relevant to the proposed study. However, detailed discussions of the research problem, related technologies, supporting literature, research gaps, and the proposed solution should be presented in the Project Context (Background of the Study).

Student Guide

The Introduction should be written as a concise narrative that provides an overview of the study and the contents of Chapter 1. It should introduce the general research domain, explain why the topic is relevant, and briefly state the purpose of the proposed study without discussing technical details or presenting the complete research problem. The discussion should gradually prepare the reader for the more comprehensive explanation found in the Project Context.

Students are encouraged to begin the Introduction by describing the broader technological or application area of the study. For example, studies involving Artificial Intelligence may briefly discuss the growing adoption of intelligent technologies across various industries, while studies on Data Analytics may introduce the increasing importance of data-driven decision-making, and Internet of Things (IoT) studies may highlight the growing use of connected devices and smart environments. The discussion should remain general and avoid focusing immediately on the specific research problem.

After introducing the research domain, briefly explain how the proposed study relates to the application area and why it is relevant. This discussion should provide enough information for readers to understand the focus of the study without presenting an extensive literature review or detailed technical explanations. When appropriate, students may include one or two recent scholarly references to support statements regarding the importance of the research domain. However, the majority of related literature should be reserved for the Project Context.



The Introduction should conclude by informing the reader about the organization of Chapter 1. Briefly mention that the chapter presents the Project Context, Statement of the Problem, Objectives of the Study, Significance of the Study, Scope and Limitations, and Definition of Terms. This provides a logical transition to the succeeding sections and helps readers understand the overall structure of the chapter.

As a general guideline, the Introduction should consist of **one to two well-developed paragraphs** containing approximately **150 to 250 words**. The writing should be formal, concise, and coherent, allowing the chapter to begin with a clear overview before presenting the detailed discussions that follow.

Writing Tips

A strong Introduction should answer the following questions naturally within the discussion:

- What is the general field or application area of the study?
- Why is this field currently important?
- What is the general purpose of the proposed research?
- What can readers expect in the remainder of Chapter 1?

The Introduction should **introduce** the study rather than **justify** it. The justification, supporting literature, research gaps, and detailed explanation of the proposed system belong in the **Project Context (Background of the Study)**.

Common Mistakes to Avoid

Students should avoid beginning the Introduction by immediately describing the research problem or proposed system. Likewise, lengthy definitions of Artificial Intelligence, Machine Learning, Data Analytics, Internet of Things, or other technologies should not be included in this section. Extensive discussions of related literature, research gaps, system architecture, methodology, or implementation details should also be avoided, as these topics are discussed in later sections of the thesis. Finally, students should refrain from copying the research objectives directly into the Introduction, since the purpose of this section is to provide an overview rather than a detailed explanation of the study.

Example 1: Artificial Intelligence, Machine Learning, and Generative AI.
“AI-Powered Student Academic Risk Prediction and Advisory System”

Artificial Intelligence (AI), Machine Learning (ML), and Generative AI have transformed the way organizations analyze information, automate decision-making, and provide intelligent services across various sectors,



including education, healthcare, business, and government. Their ability to learn from historical data, recognize complex patterns, and generate context-aware recommendations has created new opportunities for developing intelligent systems that improve efficiency and support evidence-based decision-making. As educational institutions continue to generate increasing volumes of academic and student-related data, these technologies offer promising approaches for addressing complex educational challenges and improving institutional effectiveness.

This chapter introduces the proposed AI-Powered Student Academic Risk Prediction and Advisory System, which aims to leverage Artificial Intelligence, Machine Learning, and Generative AI technologies to identify students who may be at risk of academic underperformance and provide personalized intervention recommendations. The chapter presents the Project Context, Statement of the Problem, Objectives of the Study, Significance of the Study, Scope and Limitations, and Definition of Terms, providing the foundation for understanding the proposed research and its contribution to intelligent academic decision support.

Example 2: Data Analytics and Business Intelligence. “Data Analytics-Based Enrollment Forecasting and Decision Support System”

The rapid growth of digital information has significantly increased the importance of Data Analytics and Business Intelligence in supporting organizational planning and decision-making. By transforming large volumes of data into meaningful insights, these technologies enable organizations to monitor



performance, identify trends, forecast future outcomes, and develop evidence-based strategies. Educational institutions, in particular, continuously generate academic and administrative data that can be analyzed to improve planning, optimize resource utilization, and support institutional decision-making.

This chapter introduces the proposed Data Analytics-Based Enrollment Forecasting and Decision Support System, which utilizes predictive analytics and business intelligence techniques to improve enrollment forecasting and strategic planning. The chapter provides the Project Context, Statement of the Problem, Objectives of the Study, Significance of the Study, Scope and Limitations, and Definition of Terms, establishing the foundation for the proposed research and its intended contributions to institutional decision-making.

Example 3: Internet of Things (IoT), Embedded Systems, and Smart Technologies. “Smart Campus Energy Monitoring and Predictive Control System Using IoT Sensors”

The continuous advancement of the Internet of Things (IoT), embedded systems, and smart technologies has enabled the development of intelligent environments capable of monitoring physical conditions, collecting real-time information, and supporting automated decision-making. Through interconnected sensors, communication networks, and intelligent control mechanisms, IoT technologies have been widely adopted in various applications, including smart homes, healthcare, transportation, agriculture, and educational institutions. These innovations continue to improve operational efficiency, resource utilization, and sustainability through real-time monitoring and intelligent automation.



This chapter introduces the proposed Smart Campus Energy Monitoring and Predictive Control System Using IoT Sensors, which integrates IoT technologies and predictive analytics to support intelligent campus energy management. The chapter presents the Project Context, Statement of the Problem, Objectives of the Study, Significance of the Study, Scope and Limitations, and Definition of Terms, providing the necessary foundation for understanding the proposed system and its contribution to the development of smart campus technologies.

Project Context (3-5 PAGES)

The **Project Context**, also referred to as the **Background of the Study**, establishes the rationale for conducting the research. It explains the existing problem, discusses current technologies and related studies, identifies research gaps, and concludes by presenting the proposed solution. This section should convince the reader that the research addresses a real and relevant problem and that the proposed solution is supported by existing knowledge while contributing new insights or innovations.

Unlike Chapter 2, which provides a comprehensive review of related literature, the Project Context uses selected literature to establish the research problem, justify the technologies employed, and demonstrate the need for the proposed study. Students are encouraged to cite recent, credible journal articles, conference proceedings, books, and other scholarly sources throughout the discussion rather than presenting unsupported statements.

Student Guide

The Project Context should be written as a **continuous scholarly discussion** rather than a collection of independent sections. Although the organization presented in this manual is intended to guide the writing process, the guide headings should **not** appear in the actual manuscript. Instead, students should ensure that each paragraph flows naturally into the next, creating a coherent narrative that gradually leads the reader from the research problem to the proposed solution.

Begin by introducing the application domain and describing the real-world problem that motivates the study. Explain why the problem is important, identify



the stakeholders affected, and support the discussion using recent and relevant literature. Rather than immediately discussing technologies, establish first why the problem deserves investigation.

After presenting the problem, discuss the limitations of current practices, existing systems, or conventional approaches. Explain why these approaches are no longer sufficient and identify the challenges that remain unresolved. Whenever possible, support these observations with findings from previous studies instead of relying on personal opinions.

Once the problem has been clearly established, introduce the primary technologies relevant to the proposed study. Rather than providing dictionary definitions, explain how previous researchers have applied these technologies to solve similar problems. Discuss their capabilities, advantages, and limitations while maintaining a clear connection to the research problem. Students should only discuss technologies that are directly relevant to their proposed system.

If the study incorporates additional or supporting technologies, these should be introduced after the primary technology. Explain how these technologies complement one another and contribute to the overall solution. The discussion should remain focused on the proposed study and avoid describing technologies that will not be implemented.

Following the discussion of related technologies, identify the research gap by synthesizing the reviewed literature. Explain what limitations remain in existing studies, what functionalities are still lacking, or what opportunities exist for improvement. The research gap should naturally justify why the proposed study is necessary.

Finally, conclude the Project Context by introducing the proposed system. Briefly describe its purpose, the major technologies to be integrated, and its expected contribution to the identified problem. Detailed descriptions of the methodology, system architecture, and implementation should be reserved for the succeeding chapters.

Throughout the Project Context, students should integrate related literature naturally within the discussion. Citations should support the narrative instead of appearing as isolated summaries of individual studies. The discussion should demonstrate critical understanding by connecting previous works, identifying common findings, and highlighting existing limitations that motivate the proposed research.

Example 1

Artificial Intelligence, Machine Learning, and Generative AI

Introducing the Research Problem



Student academic performance is widely recognized as one of the primary indicators of educational quality and institutional effectiveness. Higher education institutions continuously collect large amounts of academic data, including grades, attendance records, learning management system activities, and student demographic information. Despite the availability of these data, many institutions continue to rely on traditional academic monitoring approaches that identify struggling students only after poor academic performance has already occurred. Such reactive interventions often limit the ability of academic advisers to provide timely support, increasing the likelihood of course failure, delayed graduation, and student attrition. Recent studies on learning analytics emphasize that early identification of academically at-risk students enables institutions to implement proactive intervention strategies that improve student success and retention (Siemens & Baker, 2022; Viberg et al., 2020).

[Discussing Existing Approaches and Technologies](#)

Artificial Intelligence (AI) has emerged as a transformative technology capable of supporting intelligent decision-making across various educational applications. Within AI, Machine Learning enables computer systems to learn patterns from historical academic data and generate predictive models without explicit programming. Previous studies have demonstrated that classification algorithms such as Decision Trees, Random Forests, Support Vector Machines, and Artificial Neural Networks can effectively predict student performance and identify learners who may require academic intervention (Kotsiantis et al., 2004; Alyahyan & Düştögör, 2020). These predictive models allow educational



institutions to move beyond descriptive reporting by providing data-driven insights into future student performance.

Supporting Technologies

Recent developments in Generative AI have further expanded the capabilities of intelligent educational systems. Large Language Models (LLMs) can generate human-like explanations, personalized recommendations, and context-aware responses that assist educators and students in interpreting predictive results (Kasneci et al., 2023). Rather than merely informing advisers that a student belongs to a high-risk category, Generative AI can recommend intervention strategies based on the student's academic profile and learning behavior, thereby improving the usefulness of predictive analytics in academic advising.

Research Gap and Proposed Solution

Although previous studies have demonstrated the effectiveness of Machine Learning in predicting academic performance, many existing systems focus primarily on generating prediction scores or risk classifications with limited decision-support capabilities. Few systems integrate predictive Machine Learning models with Generative AI to generate personalized intervention recommendations that directly assist academic advisers. To address this limitation, the proposed AI-Powered Student Academic Risk Prediction and Advisory System integrates Machine Learning algorithms with Generative AI technologies to identify students at risk of academic underperformance and



provide intelligent, personalized intervention recommendations that support proactive academic advising and institutional decision-making.

Example 2

Data Analytics and Business Intelligence

Higher education institutions continuously generate large volumes of institutional data from admissions, enrollment, academic records, finance, and student services. These data have become valuable organizational assets that can support strategic planning and institutional decision-making when properly analyzed. However, many universities continue to rely on traditional reporting methods that summarize historical information without providing predictive insights for future planning. Manual analysis using spreadsheets and static reports often limits administrators' ability to anticipate enrollment trends, allocate resources efficiently, and make timely strategic decisions (Shmueli et al., 2020; Chen et al., 2012). The proposed system seeks to leverage Data Analytics and Business Intelligence technologies to improve enrollment forecasting and support evidence-based institutional planning.

Challenges in Traditional Enrollment Planning

Traditional enrollment planning relies heavily on historical reports, manual computations, and the experience of institutional planners. While these approaches provide descriptive information regarding previous enrollment patterns, they often lack predictive capabilities that can anticipate future changes



in student demand. According to Romero and Ventura (2020), conventional reporting systems primarily support descriptive analysis and are limited in identifying hidden patterns within institutional data. As higher education institutions become increasingly data-driven, there is a growing need for intelligent systems capable of forecasting enrollment trends and supporting proactive decision-making.

Data Analytics

Data Analytics enables organizations to transform raw data into meaningful information through statistical analysis, predictive modeling, and pattern discovery. In higher education, data analytics has been successfully applied to analyze student enrollment behavior, monitor institutional performance, evaluate resource utilization, and identify factors influencing student retention (Daniel, 2015). Predictive analytics techniques further extend these capabilities by estimating future outcomes using historical institutional data, allowing administrators to anticipate enrollment demand and prepare appropriate academic resources.

Various predictive models, including regression analysis, decision trees, random forests, and time-series forecasting algorithms, have demonstrated promising performance in predicting enrollment trends and institutional growth (Shmueli et al., 2020). These analytical approaches enable universities to move beyond descriptive reporting by generating forecasts that support strategic planning and long-term decision-making.

Business Intelligence



Business Intelligence (BI) integrates data collection, visualization, reporting, and analytical tools to support organizational decision-making. Modern BI platforms enable administrators to monitor institutional performance through interactive dashboards that summarize key performance indicators and present analytical results in an understandable format. According to Wixom and Watson (2010), Business Intelligence improves organizational performance by transforming complex datasets into actionable information for managers and decision-makers.

Interactive dashboards also enable institutional leaders to explore enrollment patterns, compare historical performance, and evaluate forecasting results in real time. When integrated with predictive analytics, Business Intelligence enhances institutional decision support by presenting forecasted enrollment trends together with visual summaries that facilitate strategic planning.

Predictive Analytics

Predictive Analytics combines statistical methods, machine learning algorithms, and historical data to estimate future events. In higher education, predictive analytics has been increasingly adopted to forecast enrollment demand, identify admission trends, and estimate resource requirements (Picciano, 2012). Compared to traditional descriptive reports, predictive models provide administrators with forward-looking information that supports proactive planning and evidence-based policy formulation.



By integrating predictive analytics into enrollment management, institutions can improve classroom allocation, faculty workload planning, budgeting, and infrastructure utilization while reducing uncertainties associated with fluctuating student populations.

Research Gap

Although numerous studies have demonstrated the effectiveness of Data Analytics and Business Intelligence in supporting institutional decision-making, many existing enrollment management systems remain focused on descriptive reporting and visualization. Few systems integrate predictive analytics with interactive Business Intelligence dashboards that enable administrators to forecast enrollment demand while simultaneously exploring analytical results through dynamic visualizations. This limitation reduces the practical usefulness of institutional data for strategic planning and policy development.

Rationale for Technology Integration

The proposed Data Analytics-Based Enrollment Forecasting and Decision Support System integrates predictive analytics and Business Intelligence technologies to address the limitations of conventional enrollment planning. Predictive models will analyze historical enrollment data to generate accurate enrollment forecasts, while Business Intelligence dashboards will visualize trends and key performance indicators that support institutional planning. Through the integration of these technologies, the proposed system aims to assist administrators in making timely, evidence-based decisions that improve resource allocation, academic planning, and institutional sustainability.



Example 3

Internet of Things (IoT), Embedded Systems, and Smart Technologies

Educational institutions consume significant amounts of electrical energy to support classrooms, laboratories, administrative offices, libraries, and other campus facilities. As universities continue expanding their physical infrastructure and technological resources, efficient energy management has become increasingly important in reducing operational costs and promoting environmental sustainability. However, many institutions continue to rely on conventional utility monitoring practices that provide only periodic reports of electricity consumption, making it difficult to identify abnormal usage patterns and respond promptly to energy-related issues (Atzori et al., 2017; Gubbi et al., 2013). The proposed study seeks to develop an intelligent energy monitoring system that integrates Internet of Things (IoT) technologies and predictive analytics to improve campus energy management.

Challenges in Traditional Energy Monitoring

Traditional energy management systems often depend on manual meter readings and periodic utility reports that provide limited visibility into real-time energy consumption. These approaches delay the detection of equipment malfunction, energy wastage, and abnormal consumption patterns, reducing opportunities for preventive maintenance and timely intervention. According to Al-Fuqaha et al. (2015), conventional monitoring approaches lack the continuous data collection necessary for intelligent resource management and automated decision-making.



Internet of Things (IoT)

The Internet of Things (IoT) enables physical devices equipped with sensors, communication modules, and embedded processors to collect and exchange information through interconnected networks. IoT technologies have become fundamental components of smart environments by enabling continuous monitoring of environmental conditions and operational activities (Atzori et al., 2017). Within educational institutions, IoT sensors can continuously monitor electricity consumption, environmental conditions, equipment utilization, and facility operations, providing administrators with real-time information for operational management.

Recent studies have demonstrated that IoT-based monitoring systems improve operational efficiency by providing automated data collection, remote accessibility, and immediate detection of abnormal conditions (Gubbi et al., 2013). These capabilities enable institutions to reduce manual monitoring while improving the accuracy and timeliness of energy management decisions.

Embedded Systems

Embedded systems serve as the computational foundation of IoT devices by controlling sensors, collecting measurements, and processing information before transmitting data to central servers. Microcontrollers such as Arduino, ESP32, and Raspberry Pi have become widely adopted platforms for developing intelligent monitoring applications because of their affordability, flexibility, and compatibility with wireless communication technologies (Monk, 2022).



The integration of embedded systems enables continuous acquisition of electrical measurements while supporting local processing and intelligent device control. These capabilities improve system responsiveness and reduce communication delays in real-time monitoring applications.

Cloud Computing and Predictive Analytics

Cloud computing provides scalable infrastructure for storing, processing, and analyzing sensor-generated data collected by IoT devices. Through cloud-based platforms, administrators can remotely access monitoring dashboards, historical consumption records, and system alerts from various locations. Predictive analytics further enhances IoT monitoring by analyzing historical energy consumption patterns to forecast future electricity demand and identify potential equipment failures before they occur.

Recent studies indicate that integrating predictive analytics with IoT monitoring significantly improves energy optimization, preventive maintenance, and intelligent facility management (Borgia, 2014). These technologies enable organizations to move beyond reactive monitoring toward proactive resource management.

Research Gap

Although many IoT-based monitoring systems successfully provide real-time visualization of energy consumption, existing implementations primarily emphasize monitoring and reporting functionalities. Limited studies have integrated predictive analytics capable of forecasting future energy consumption and generating intelligent recommendations for campus energy optimization.



Consequently, facility managers continue to rely on manual interpretation of monitoring data when making operational decisions.

Rationale for Technology Integration

The proposed Smart Campus Energy Monitoring and Predictive Control System Using IoT Sensors integrates Internet of Things technologies, embedded systems, cloud computing, and predictive analytics to provide intelligent energy management for educational institutions. IoT sensors and embedded devices will continuously collect real-time energy consumption data, while predictive analytics will forecast future energy demand and identify abnormal consumption patterns. Cloud-based dashboards will provide administrators with centralized access to monitoring information and intelligent recommendations, enabling proactive decision-making that promotes operational efficiency, cost reduction, and sustainable campus management.

Objectives of the Study

The General Objective should present the overall purpose of the study in a short narrative rather than as a single isolated statement. It should begin with a phrase such as “The main objective of this study is to...” followed by the primary action or intended research output. The succeeding sentence should briefly explain what the proposed system, model, or application is expected to accomplish and identify the principal technologies or approaches involved.

The General Objective should provide an overview of the complete study without listing the detailed activities, algorithms, features, or evaluation metrics. These details should be presented in the Specific Objectives.

The General Objective should then transition naturally to the Specific Objectives using a statement such as:

“The study specifically aims to address the following objectives:”

For system-development research, the four Specific Objectives should generally follow this sequence:



Data → Algorithms/Models → System/Application → Evaluation

The first objective should address the identification, collection, preparation, or preprocessing of the data required by the study. The second should address the implementation, development, or comparison of the algorithms or models. The third should address the design and development of the proposed system or application. The fourth should evaluate **both the algorithms/models and the developed system**, using metrics appropriate to each.

The exact algorithms and evaluation metrics should be selected according to the research problem. For example, classification studies may use accuracy, precision, recall, F1-score, specificity, ROC-AUC, and confusion matrices, while regression studies may use MAE, MSE, RMSE, and R^2 . Clustering studies may use measures such as silhouette score, Davies–Bouldin Index, or Calinski–Harabasz Index. For software evaluation, students may use an established framework such as ISO/IEC 25010 or a usability instrument such as the System Usability Scale (SUS), provided that the selected instrument is appropriate for the study.

Specific Objective 1 — Data

The first Specific Objective should establish the **data foundation of the study**. It should identify the type, source, and relevant characteristics of the data required by the algorithms and system. Where appropriate, the objective may also specify the preparation or preprocessing activities, such as cleaning, transformation, feature selection, normalization, annotation, or labeling.

Students should be specific enough to demonstrate that the proposed dataset can support the research objectives. However, the objective should not become a detailed data-management procedure. Details such as the exact preprocessing steps, train-test split, missing-value treatment, and feature engineering should normally be explained in Chapter 3.

Specific Objective 2 — Algorithms or Models

The second Specific Objective should identify the computational techniques that will be developed, implemented, or compared. When the research already has a defined methodological direction, students may explicitly name the candidate algorithms.

For example, a classification study may compare Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, and XGBoost. An NLP study may investigate BERT, RoBERTa, or traditional TF-IDF-based classifiers. A forecasting study may compare SARIMA, Random Forest Regression, XGBoost, LSTM, or hybrid models, depending on the research design.



The objective should also make clear whether the algorithms will be implemented, trained, optimized, or compared.

Specific Objective 3 — System/Application

The third Specific Objective should focus on the **actual software, application, or intelligent system** that will be developed. It should show how the selected algorithm or model will be integrated into a functional solution.

Students may mention the major functions of the application, such as prediction, recommendation, visualization, conversational interaction, monitoring, reporting, or decision support. They do not need to enumerate every interface element.

Specific Objective 4 — Evaluation

The fourth Specific Objective should evaluate **two different dimensions of the research. First, the algorithm/model must be evaluated to determine whether it performs the computational task effectively. Second, the developed system must be evaluated to determine whether it functions effectively and is usable by its intended users.** These should not be treated as the same evaluation.

For classification, appropriate metrics may include:

- Accuracy
- Precision
- Recall
- F1-score
- Specificity
- ROC-AUC
- Confusion Matrix

For regression or forecasting:

- MAE
- MSE
- RMSE
- MAPE
- R^2

For clustering:

- Silhouette Score
- Davies–Bouldin Index
- Calinski–Harabasz Index

For NLP generation or retrieval, depending on the task:

- BLEU



- ROUGE
- BERTScore
- Precision@K
- Recall@K
- Mean Reciprocal Rank (MRR)

For system evaluation, students may use:

- **ISO/IEC 25010** software quality characteristics;
- **System Usability Scale (SUS)** for usability;
- task completion rate;
- response time;
- error rate;
- user satisfaction.

Students should **not automatically use all of these metrics**. They should select metrics appropriate to the actual research problem and methodology.

Example 1: Artificial Intelligence, Machine Learning, and Generative AI

Objectives of the Study

The main objective of this study is to develop and implement an AI-powered student academic risk prediction and advisory system that integrates Machine Learning and Generative AI technologies. The system aims to identify students who may be at risk of academic underperformance and provide personalized intervention recommendations to support timely, data-driven academic decision-making. The study specifically aims to address the following objectives:

1. To identify, collect, and preprocess relevant student academic data, including grades, attendance, assessment results, and other academic indicators, required for predicting student academic risk;
2. To implement and compare appropriate Machine Learning classification algorithms, such as Logistic Regression, Decision Tree, Random Forest,



Support Vector Machine, and XGBoost, for classifying students according to their level of academic risk using accuracy, precision, recall, F1-score, and confusion matrix;

3. To design and develop an AI-powered student academic risk prediction and advisory system that integrates the selected Machine Learning model with Generative AI to provide personalized and contextually appropriate academic intervention recommendations; and
4. To evaluate the performance of the implemented Machine Learning algorithms in terms of accuracy, precision, recall, F1-score, and confusion matrix, and evaluate the developed system using applicable ISO/IEC 25010 software quality characteristics and the System Usability Scale (SUS).

Example 2: Data Analytics and Business Intelligence

Objectives of the Study

The main objective of this study is to develop and implement a data analytics-based enrollment forecasting and decision support system that integrates predictive analytics and Business Intelligence technologies. The system aims to analyze historical enrollment patterns, generate reliable enrollment forecasts, and provide interactive analytical information to support evidence-based institutional planning and decision-making. The study specifically aims to address the following objectives:



1. To identify, collect, and preprocess historical enrollment data, including enrollment counts by academic year, academic program, year level, and other relevant institutional variables, required for enrollment forecasting;
2. To implement and compare appropriate forecasting algorithms or models, such as Seasonal ARIMA (SARIMA), Random Forest Regression, and Long Short-Term Memory (LSTM), for predicting future enrollment trends using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE);
3. To design and develop a data analytics-based enrollment forecasting and decision support system that integrates the selected forecasting model with interactive Business Intelligence dashboards for presenting historical enrollment trends, forecasts, and relevant decision-support information; and
4. To evaluate the performance of the implemented forecasting algorithms or models in terms of MAE, RMSE, and MAPE, and evaluate the developed system using applicable ISO/IEC 25010 software quality characteristics and the System Usability Scale (SUS).

Example 3: Internet of Things, Embedded Systems, and Smart Technologies

Objectives of the Study

The main objective of this study is to develop and implement a smart campus energy monitoring and predictive control system that integrates Internet of Things (IoT), embedded systems, and predictive analytics technologies. The system aims to provide real-time monitoring of energy consumption, forecast



future energy demand, and identify abnormal energy usage patterns to support more efficient and informed campus energy management. The study specifically aims to address the following objectives:

1. To identify, collect, and preprocess real-time and historical energy consumption data, including electricity consumption, voltage, current, power, and other relevant sensor measurements, required for energy monitoring and predictive analysis;
2. To implement and compare appropriate predictive algorithms or models, such as Random Forest Regression, XGBoost, and Long Short-Term Memory (LSTM), for forecasting energy consumption using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE);
3. To design and develop an IoT-based smart campus energy monitoring and predictive control system that integrates energy sensors, embedded devices, the selected predictive model, data storage, and a monitoring dashboard; and
4. To evaluate the performance of the implemented predictive algorithms or models in terms of MAE, RMSE, and MAPE, and evaluate the developed system in terms of functionality, performance efficiency, reliability, and usability using applicable ISO/IEC 25010 software quality characteristics and the System Usability Scale (SUS).

Significance of the Research (arranged based on importance, at-least 5, including researchers and future researchers)

Purpose



The Significance of the Research explains the potential value, contribution, and practical relevance of the study to specific stakeholders. It should answer the question:

Who can benefit from the study, and in what specific way can the results, system, model, or findings be useful to them?

The section should identify the groups that are most directly affected by the research and explain how each may benefit from the proposed system or findings. The discussion should be based on the actual outputs and intended users of the study rather than making broad or unsupported claims.

The beneficiaries should generally be arranged from those who are expected to receive the most direct and immediate benefit to those who may benefit from the research in a broader or longer-term context. At least five stakeholder groups should be discussed, and Researchers and Future Researchers should be included whenever appropriate.

Student Guide

Begin the Significance of the Research by briefly explaining the overall contribution of the study. The discussion should then identify the primary stakeholder or intended users of the proposed system and explain how the system can address their needs, improve their existing practices, or support their decision-making.

After discussing the primary beneficiaries, identify other stakeholders who may benefit from the research. These may include administrators, faculty members, students, technical personnel, industry practitioners, organizations, policymakers, researchers, or other groups depending on the nature of the study. Students should select beneficiaries based on the **actual users and potential applications of their research**, rather than simply listing common stakeholder categories.

Each beneficiary should be presented using a **descriptive subheading followed by a paragraph explaining the specific benefit**. Although these subheadings are useful for organizing the discussion, the actual content should remain in paragraph form rather than being presented as bullet points.

The benefits should be specific and realistic. For example, instead of stating that *“the system will benefit administrators by improving efficiency,”* explain **which administrative activity will become more efficient and how the system contributes to that improvement**. Similarly, instead of saying that *“future researchers can use this study,”* explain what they may actually use, such



as the dataset, methodology, algorithm comparison, system architecture, evaluation results, or identified limitations.

Researchers should also distinguish between **direct benefits** and **potential contributions**. A proposed system may provide direct operational benefits to its intended users, while researchers and future researchers may benefit primarily from the knowledge, methodology, datasets, models, findings, or research opportunities generated by the study.

The final beneficiary should generally be **Future Researchers**, emphasizing how the study can serve as a reference or starting point for improving the proposed approach, testing alternative algorithms, applying the technology to other domains, or addressing limitations identified by the researchers.

Example 1: Artificial Intelligence, Machine Learning, and Generative AI

Significance of the Research

The proposed AI-powered student academic risk prediction and advisory system may support early identification of students who are at risk of academic underperformance and provide data-driven recommendations for academic intervention. The study may benefit the following stakeholders:

Students. Students may benefit from earlier identification of academic risks and personalized intervention recommendations. These recommendations may help them recognize areas requiring attention and seek appropriate academic support.

Academic Advisers and Faculty Members. The system may help advisers and faculty members identify students who require academic intervention. The predicted risk levels and generated recommendations may provide additional information for planning appropriate support strategies.



Academic Administrators. Administrators may use aggregated academic risk information to identify common areas of concern and support the development of appropriate academic intervention programs.

AI and Software Developers. The study may provide developers with a practical reference for integrating Machine Learning-based prediction and Generative AI-based recommendations into an educational application.

Researchers. The researchers may gain experience in academic data preparation, Machine Learning model development, Generative AI integration, and intelligent system evaluation. The study may also provide findings on the performance of selected prediction models.

Future Researchers. Future researchers may use the study as a reference for testing other Machine Learning algorithms, Generative AI techniques, predictors, or approaches to academic risk prediction and advisory systems.

Example 2: Data Analytics and Business Intelligence

The proposed data analytics-based enrollment forecasting and decision support system may support institutional planning by providing enrollment forecasts and interactive analytical information. The study may benefit the following stakeholders:

Institutional Administrators. Administrators may use enrollment forecasts to support decisions involving academic programs, staffing,



classrooms, and other institutional resources. The system may provide additional evidence for planning future enrollment requirements.

Enrollment and Planning Personnel. Enrollment and planning personnel may use the system to analyze historical enrollment patterns and generate forecasts. This may reduce the effort required to manually consolidate and interpret enrollment information.

Academic Departments. Academic departments may use enrollment projections to support planning for class sizes, course offerings, and faculty requirements. The forecasts may provide additional information for departmental planning.

Business Intelligence and Data Analytics Professionals. The study may provide practitioners with a practical example of integrating predictive analytics with Business Intelligence dashboards. The approach may serve as a reference for similar data-driven decision-support applications.

Researchers. The researchers may gain experience in data preparation, forecasting model development, model comparison, Business Intelligence visualization, and system evaluation. The study may also provide comparative findings on the selected forecasting models.

Future Researchers. Future researchers may extend the study by incorporating additional enrollment variables, testing other forecasting algorithms, or applying the approach to other institutional forecasting problems such as retention or resource demand.

Example 3: Internet of Things, Embedded Systems, and Smart Technologies



The proposed smart campus energy monitoring and predictive control system may support more efficient campus energy management through real-time monitoring and energy consumption forecasting. The study may benefit the following stakeholders:

Facility and Energy Management Personnel. Personnel may use real-time energy information to monitor consumption and identify unusual usage patterns. This may support faster investigation and more informed energy management decisions.

Campus Administrators. Administrators may use energy consumption trends and forecasts to support decisions related to energy conservation and facility management. The information may also support institutional sustainability initiatives.

Maintenance Personnel. Maintenance personnel may use abnormal energy consumption information to identify equipment or operational conditions that require inspection. This may support preventive maintenance activities.

IoT and Embedded Systems Developers. The study may provide developers with a practical reference for integrating sensors, embedded devices, predictive models, and monitoring dashboards into an IoT-based application.

Researchers. The researchers may gain experience in IoT development, sensor data acquisition, predictive modeling, system integration, and system evaluation. The study may also provide findings regarding the performance of selected predictive models using energy consumption data.



Future Researchers. Future researchers may extend the study by testing other predictive algorithms, adding more sensors, or applying the approach to other smart-campus applications such as water monitoring, environmental monitoring, or facility management.

Scope and Limitations of the Project (discussions should be aligned with your objectives)

Student Guide

The **Scope and Limitations of the Project** should define the boundaries of the study and should be directly aligned with the Specific Objectives. The scope should explain the **data to be used, algorithms or models to be implemented, system or application to be developed, intended users, and evaluation procedures** covered by the project.

The scope should generally be presented in several paragraphs following the sequence of the Specific Objectives: **Data → Algorithms/Models → System/Application → Evaluation**. Each paragraph should describe only the components that are included in the study.

The **limitations should be presented in the final paragraph**. This paragraph should identify the major constraints, conditions, or exclusions that may affect the implementation, results, or applicability of the study. These may include limitations related to data availability and quality, algorithm or model performance, hardware or software resources, number of users, deployment environment, implementation period, or other factors specific to the project.

Students should avoid introducing major features or technologies in the Scope that are not included in the objectives. Similarly, limitations should not simply list features that were intentionally excluded unless their absence has a meaningful effect on the study.

Example 1: Artificial Intelligence, Machine Learning, and Generative AI

Scope and Limitations of the Project

The scope of this study focuses on the design, development, and implementation of an AI-powered student academic risk prediction and advisory system that integrates Machine Learning and Generative AI technologies. The



project will focus on using academic data to predict students who may be at risk of academic underperformance and providing personalized intervention recommendations to support academic decision-making.

The data component of the study will include relevant student academic information, such as grades, attendance, assessment results, and other available academic indicators associated with academic performance. The data will be collected, cleaned, transformed, and prepared for Machine Learning model development. Only data that are available and authorized for use in the study will be included.

The algorithm component will focus on selected Machine Learning classification algorithms, such as Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, and XGBoost. These algorithms will be implemented and compared to determine their capability to classify students according to academic risk using appropriate performance measures, including accuracy, precision, recall, F1-score, and confusion matrix. The selected model will subsequently be integrated into the proposed system.

The system component will include functionalities for authorized users to access relevant academic information, generate student academic risk predictions, view prediction results, and obtain Generative AI-based intervention recommendations. The system will be designed as a decision-support application for academic advisers, faculty members, or other authorized personnel involved in student monitoring and intervention.



The evaluation component will cover both the Machine Learning models and the developed system. The models will be evaluated using accuracy, precision, recall, F1-score, and confusion matrix, while the developed system will be evaluated using applicable ISO/IEC 25010 software quality characteristics and the System Usability Scale (SUS). User evaluation will involve selected authorized users who will assess the functionality and usability of the developed application.

The study is limited by the availability, completeness, and quality of the academic data used for model development. The performance of the predictive models may be affected by limited data, imbalanced classes, incomplete records, and academic factors that are not represented in the dataset. The study will also be limited to the selected Machine Learning algorithms and Generative AI approach and will not attempt to identify all possible causes of academic underperformance. Furthermore, the system will serve only as a decision-support tool, and its predictions and recommendations will not replace the professional judgment of faculty members or academic advisers.

Example 2 — Data Analytics and Business Intelligence

The scope of this study focuses on the design, development, and implementation of a data analytics-based enrollment forecasting and decision support system that integrates predictive analytics and Business Intelligence technologies. The project will analyze historical enrollment patterns, generate enrollment forecasts, and present analytical information to support institutional planning and decision-making.



The data component of the study will include historical enrollment information such as enrollment counts, academic year, academic program, year level, and other relevant institutional variables available for research purposes. The data will be collected, cleaned, transformed, and prepared for forecasting. Only relevant and authorized institutional data within the defined study period will be included.

The algorithm component will focus on selected forecasting models, such as Seasonal ARIMA (SARIMA), Random Forest Regression, and Long Short-Term Memory (LSTM). The models will be implemented and compared to determine their capability to forecast future enrollment trends using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE). The selected model based on the defined evaluation criteria will be considered for integration into the proposed system.

The system component will include functionalities for presenting historical enrollment trends, generating enrollment forecasts, and displaying relevant analytical information through interactive Business Intelligence dashboards. The system will be designed for authorized institutional administrators, planning personnel, and academic departments who require enrollment information for planning and decision-making.

The evaluation component will cover both the forecasting models and the developed system. The forecasting models will be evaluated using MAE, RMSE, and MAPE, while the developed system will be evaluated using applicable ISO/IEC 25010 software quality characteristics and the System Usability Scale



(SUS). Selected institutional personnel will participate in the evaluation of the system's functionality and usability.

The study is limited by the availability and quality of historical enrollment data and the variables included in the dataset. Forecasting performance may be affected by unexpected events, changes in enrollment policies, changes in academic programs, and other external factors that are not represented in the historical data. The study will focus only on enrollment forecasting and related decision-support functions and will not cover other institutional forecasting applications, such as student retention, financial forecasting, or faculty workload prediction, unless specifically included in the research design.

Example 3 — Internet of Things, Embedded Systems, and Smart Technologies

The scope of this study focuses on the design, development, and implementation of an IoT-based smart campus energy monitoring and predictive control system. The project will integrate IoT sensors, embedded devices, predictive analytics, and a monitoring application to provide real-time energy information and forecast energy consumption for selected campus facilities.

The data component of the study will include real-time and historical energy measurements, such as electricity consumption, voltage, current, power, and other relevant sensor readings. The data will be collected through selected IoT sensors and embedded devices and will undergo appropriate cleaning, transformation, and preprocessing before being used for predictive analysis.

The algorithm component will focus on selected predictive models, such as Random Forest Regression, XGBoost, and Long Short-Term Memory (LSTM),



for forecasting energy consumption. The models will be implemented and compared using appropriate forecasting metrics, including Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE). Where applicable, the selected model may also be used to identify abnormal energy consumption patterns.

The system component will integrate the selected IoT sensors, embedded devices, communication mechanisms, data storage, predictive model, and monitoring dashboard. The developed system will provide authorized users with access to real-time or near-real-time energy consumption information, historical trends, forecasts, and other relevant indicators for campus energy management.

The evaluation component will cover both the predictive models and the developed IoT system. The predictive models will be evaluated using MAE, RMSE, and MAPE, while the developed system will be evaluated using applicable ISO/IEC 25010 software quality characteristics, particularly those relevant to functionality, performance efficiency, reliability, and usability. The System Usability Scale (SUS) may also be used to assess the perceived usability of the developed application.

The study is limited by the number, location, accuracy, and capabilities of the available IoT sensors and embedded devices. Sensor errors, connectivity interruptions, power availability, network conditions, and missing readings may affect the quality and continuity of the collected data and consequently influence model performance. The study will also be limited to selected campus facilities and monitored energy parameters during the defined implementation period and



will not cover all campus buildings or all forms of resource consumption. The predictive component will provide forecasts and analytical information but will not automatically control electrical equipment unless such automated control is explicitly included in the system design.

Definition of Terms

The Definition of Terms section provides operational definitions of important terms used in the study. The purpose is to ensure that readers understand how key concepts, technologies, variables, algorithms, and system components are specifically interpreted within the context of the research.

Students should identify terms primarily from the research title, General Objective, Specific Objectives, major system components, and key concepts discussed in Chapter 1. Not every technical term appearing in the paper needs to be defined. Terms that are commonly understood and do not have a study-specific meaning may be omitted.

Each term should be defined operationally. This means that students should explain what the term means as it is used or applied in their study, rather than simply copying a dictionary, textbook, or general technical definition.

Definitions should be short, clear, and precise, generally limited to one or two sentences. The definition should identify the role or meaning of the term within the study. For example, instead of defining *Machine Learning* generally as “a branch of Artificial Intelligence that enables computers to learn from data,” a study may define it as “the computational approach used in this study to develop models for classifying students according to academic risk.”

The terms may be arranged alphabetically for easier reference. Abbreviations should be written in full when first introduced and may be included in the Definition of Terms when they are central to the study.

Recommended Selection of Terms

Students may select terms from the following categories:

- Key technologies — Artificial Intelligence, Machine Learning, Generative AI, IoT, Business Intelligence
- Data-related terms — Dataset, Data Preprocessing, Data Requirements
- Algorithms/models — Random Forest, XGBoost, LSTM, SVM
- System components — Decision Support System, Dashboard, Chatbot



- Research variables/concepts — Academic Risk, Enrollment Forecasting, Energy Consumption
- Evaluation concepts — F1-score, RMSE, SUS, ISO/IEC 25010
- Important domain-specific terms appearing in the title or objectives

Students should not define every algorithm, software library, programming language, or common technical term unless it is central to understanding the study.

Example 1 — Artificial Intelligence, Machine Learning, and Generative AI

Definition of Terms

The following terms are operationally defined based on their use in the proposed study:

Academic Risk – The predicted likelihood that a student may experience academic underperformance based on the academic indicators included in the study.

Artificial Intelligence (AI) – The computational technology used in this study to support student academic risk prediction and generate personalized academic intervention recommendations.

Generative AI – The AI technology integrated into the proposed system to generate personalized academic intervention recommendations based on the student's predicted academic risk and available academic information.

Machine Learning (ML) – The computational approach used in this study to develop models that classify students according to their level of academic risk based on available academic data.

Student Academic Risk Prediction – The process of using Machine Learning models to classify students according to their likelihood of experiencing academic underperformance.



Academic Intervention Recommendation – The personalized recommendation generated by the system to provide possible academic support strategies based on the predicted risk level of a student.

F1-Score – The classification performance measure used in this study to determine the balance between precision and recall of the implemented Machine Learning models.

System Usability Scale (SUS) – The usability assessment instrument used in this study to measure users' perceived usability of the developed academic risk prediction and advisory system.

Example 2 — Data Analytics and Business Intelligence

Definition of Terms

The following terms are operationally defined based on their use in the proposed study:

Business Intelligence (BI) – The technology used in this study to present enrollment trends, forecasts, and analytical information through interactive dashboards for institutional decision-making.

Data Analytics – The process used in this study to examine historical enrollment data and generate information that supports enrollment forecasting and institutional planning.

Decision Support System (DSS) – The developed application that presents enrollment forecasts and analytical information to assist authorized institutional personnel in planning and decision-making.



Enrollment Forecasting – The process of predicting future enrollment levels using historical enrollment data and selected forecasting models.

Long Short-Term Memory (LSTM) – A recurrent neural network model implemented in this study as one of the candidate approaches for forecasting future enrollment trends.

Mean Absolute Error (MAE) – The forecasting evaluation metric used in this study to measure the average absolute difference between actual and predicted enrollment values.

Mean Absolute Percentage Error (MAPE) – The forecasting evaluation metric used to express the average prediction error as a percentage of the actual enrollment values.

Root Mean Squared Error (RMSE) – The forecasting evaluation metric used in this study to measure the magnitude of prediction errors while giving greater weight to larger errors.

Seasonal ARIMA (SARIMA) – A time-series forecasting model implemented in this study as one of the candidate models for predicting future enrollment trends.

Example 3 — Internet of Things, Embedded Systems, and Smart Technologies

Definition of Terms

The following terms are operationally defined based on their use in the proposed study:

Energy Consumption – The amount of electrical energy measured by the selected sensors and used in this study for monitoring and predictive analysis.



Internet of Things (IoT) – The technology framework used in this study to collect and transmit energy-related measurements from sensors installed in selected campus facilities.

Long Short-Term Memory (LSTM) – A recurrent neural network model implemented in this study as one of the candidate approaches for forecasting campus energy consumption.

Predictive Model – The computational model used in this study to estimate future energy consumption based on collected historical and real-time data.

Random Forest Regression – A machine learning regression algorithm implemented in this study as one of the candidate models for predicting energy consumption.

Smart Campus – The campus environment covered by the proposed system in which IoT sensors and digital technologies are used to monitor and analyze energy consumption.

Smart Energy Monitoring System – The developed application that collects, processes, and presents energy consumption information from selected campus facilities.

System Usability Scale (SUS) – The usability assessment instrument used in this study to measure users' perceived usability of the developed energy monitoring system.

XGBoost – A gradient-boosting machine learning algorithm implemented in this study as one of the candidate models for forecasting energy consumption.

Important Rule: Define Terms Based on the Actual Study



✗ Too General

Machine Learning – A branch of Artificial Intelligence that enables computers to learn from data without being explicitly programmed.

This is technically correct, but it does not tell the reader how Machine Learning is used in the study.

✓ Operational

Machine Learning (ML) – The computational approach used in this study to develop models that classify students according to their level of academic risk based on available academic data.

The second definition is better because it connects the term directly to the research objective and application.

The terms should correspond to what actually appears in the objectives.

For example, if Objective 2 says:

“To implement and compare Random Forest, SVM, and XGBoost...” then those algorithms can appropriately appear in the Definition of Terms.

But if the final study does not use SVM, students should remove SVM from the Definition of Terms rather than defining an algorithm that is no longer part of the research. Likewise, if Objective 4 specifies F1-score and SUS, those terms can be operationally defined because they are part of the study's evaluation framework.

Remove all Notes and place it as Bibliography (alphabetically arrange using APA format)